

A counterexample to canonical Toeplitz eigenvalue distribution

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Abstract

We construct a computably specified continuous complex scalar symbol with neither an inner nor an outer annular holomorphic extension, whose Toeplitz sections are not canonically distributed. Along an infinite sequence of orders, each of two points outside the symbol range is an eigenvalue of fixed positive asymptotic multiplicity. The construction uses exact finite spectral corrections, geometric Cauchy coordinates, and a uniformly convergent Fourier series. A deterministic search using finite contraction certificates specifies every correction. All parameter choices are explicit, with no claim of computational efficiency.

1 Statement and notation

For a continuous complex scalar symbol a on $\mathbb{T}$, write

$$a_k = \frac{1}{2\pi} \int_0^{2\pi} a(e^{it}) e^{-ikt} dt, \quad T_n(a) = (a_{j-k})_{j,k=0}^{n-1}.$$

The canonical-distribution conjecture, in the continuous-symbol formulation recorded in [1, Section 1, p. 2], asserts convergence of the empirical eigenvalue measures to $a_* m$ when both separate one-sided annular extensions are absent. The problem is catalogued as OP-00159 at [clopén.solutions](#) [3] and as SP-14 in [OpenProblemsInNLA](#) [4]. Theorem 6.1 gives a counterexample under exactly these hypotheses. Its symbol has the form

$$a(z) = zg(z^2), \quad g_0(s) = \sqrt{1+s^{-1}}, \quad g = g_0 + P_- + P_+.$$

Every finite background P_- has strictly negative powers and P_+ has strictly positive powers; each separately vanishes at -1 . The square root is the branch analytic outside the unit circle, equal to one at infinity and continuous on the circle. Set $a_0(z) = zg_0(z^2)$ and $h(s) = sg(s)^2 - 1$. Smallness of $h - s$ in the norms below makes $h(\mathbb{T})$ a Jordan curve with bounded interior D containing zero. Its normalized conformal map is $f : \mathbb{D} \rightarrow D$, $f(0) = 0$, $f'(0) > 0$; real symmetry gives $f(-1) = -1$.

For Laurent series and analytic series, respectively, use

$$\|v\|_{\mathcal{W}^\beta} = \sum_{j \in \mathbb{Z}} (1 + |j|)^\beta |v_j|, \quad \|x\|_{H^s}^2 = \sum_{k \geq 0} (k+1)^{2s} |x_k|^2.$$

The two-sided version of H^s uses $1 + |j|$. Write $P_n x = \sum_{k=0}^{n-1} x_k z^k$. Write Π_+, Π_- for the nonnegative and strictly negative Fourier projections, and $T(g_0)V = \Pi_+(g_0V)$ on analytic polynomials. The background polynomials remain denoted by P_-, P_+ .

2 Exact finite algebra and the boundary inverse estimates

Proposition 2.1 (Odd-symbol pencil and its jets). *Let $a(z) = zg(z^2)$, $n = 2m + 1$, $L = m + 1$, and let U be the L -by- L upper unit shift. Put*

$$G = T_L(g), \quad H = G^2 - U, \quad M(t) = e_0^*(H - tU)^{-1}e_m.$$

*Whenever H is invertible, divisibility of $\det(wI - T_n(a))/w$ by $(w^2 - 1)^h$ is equivalent to $[t^k]M(t) = 0$ for $0 \leq k < h$. This is an exact equivalence of finite algebraic constraints. If $W = H^{-1}$ and $r_k = e_0^*W(UW)^k$, then*

$$[t^k]M(t) = (r_k)_m.$$

Write $R_k(s) = \sum_{j=0}^m (r_k)_j s^j$ and $X_k(s) = \sum_{j=0}^m (r_k G)_j s^j$. For a change v_q in g_{-q-1} ,

$$\frac{\partial [t^k]M(t)}{\partial v_q} = -2 \sum_{l=0}^k [s^{m-q-1}] R_l(s) X_{k-l}(s). \quad (1)$$

At $g(s) = \sqrt{1 + s^{-1}}$ one has $H = I$, $M(t) = t^m$, and

$$J_{kq}^0 = -2(k+1) \binom{1/2}{m-q-1-k} \mathbf{1}_{k \leq m-q-1}.$$

Finally, if $\|g - \sqrt{1 + s^{-1}}\|_\infty \leq \eta$, then

$$\|H - I\| \leq 2\sqrt{2}\eta + \eta^2.$$

Thus sufficiently small symbol perturbations give invertibility of H uniformly in m , without an asymptotic finite-section argument.

Proof. Let $c_w(z) = z^{-1}(a(z) - w)$ and order the even coordinates before the odd ones in $T_{2m+2}(c_w)$. Its block form is

$$\begin{pmatrix} G & -wI \\ -wU & G \end{pmatrix}.$$

Its determinant is $\det(G^2 - w^2U)$: first use the Schur complement when G is invertible, then use the polynomial identity to remove that restriction. For $w \neq 0$, solving for the first inverse row gives the two blocks

$$e_0^*(G^2 - w^2U)^{-1}G, \quad we_0^*(G^2 - w^2U)^{-1}.$$

Jacobi's identity at the upper-right corner gives

$$\det(wI - T_n(a)) = w \det(H - tU)M(t), \quad t = w^2 - 1.$$

The two signs cancel because n is odd. Since $\det H \neq 0$, the factor multiplying M is nonzero at $t = 0$. Oddness of the characteristic polynomial supplies the matching multiplicity at $w = -1$. The resolvent series proves the row-jet formula.

All Toeplitz matrices here are persymmetric with respect to the reversal matrix J_L . So are $H - tU$ and its inverse, and each coefficient $W(UW)^k$. Their last columns are the reversals of their first rows, with a transpose and no complex conjugation. Changing g_{-q-1} changes G by $E = U^{q+1}$ and changes H to first order by $EG + GE$. Differentiating the resolvent gives

$$-\sum_{l=0}^k r_l (EG + GE) J_L r_{k-l}^\top.$$

The two sums agree after reversing l and $k-l$. The remaining matrix product selects the coefficient of degree $m-q-1$, proving (1). At the base symbol, $G = \sqrt{I+U}$, so $G^2 - U = I$; the displayed triangular formula follows. The multiplier bound $\|\sqrt{I+U}\| \leq \sqrt{2}$ and $\|G - G_0\| \leq \eta$ prove the last assertion. $\square$

Lemma 2.2 (Changing the spectral coordinate exactly). *Let $f(u) = \sum_{l \geq 1} f_l u^l$ be analytic near zero with $f_1 \neq 0$, and hold this function fixed while varying G . The first h jets of $e_0^*(H - f(u)U)^{-1}e_m$ vanish if and only if the first h jets of $M(t)$ vanish. Set $W = H^{-1}$ and define rows*

$$r_0 = e_0^*W, \quad r_k = \left(\sum_{l=1}^k f_l r_{k-l} \right) UW.$$

Then r_k is the k th row coefficient of the transformed inverse. The Jacobian formula (1) remains valid with these rows and their products with G .

Proof. Composition by f acts on the first h Taylor coefficients through a lower triangular matrix with diagonal $1, f_1, \dots, f_1^{h-1}$. This matrix is invertible. The resolvent identity gives the row recursion, and differentiation and persymmetry give the unchanged bilinear formula. Fixing f avoids extra derivatives from the coordinate choice; an adaptive coordinate would require accounting for them. $\square$

Lemma 2.3 (Suppression of repeated corner corrections). *Suppose $G = T_{m+1}(g)$ has lower bandwidth p , and set $H = G^2 - U$. Let $h + p < (m+2)/2$, $Q = m - h$, and $E = \sum_{q=Q}^{m-1} v_q U^{q+1}$. Then $E^2 = 0$ and*

$$(G + E)^2 - U = H + D, \quad D = GE + EG.$$

Let A select coordinates $0, \dots, h + p - 1$, and let B select $m + 1 - h - p, \dots, m$. Then $D = ADB$. Suppose, for $|t| \leq R$, that

$$\|(H - tU)^{-1}\| \leq M_0, \quad \|D_r(H - tU)^{-1}D_r^{-1}\| \leq M_r, \quad D_r = \text{diag}(1, r, \dots, r^m), \quad r > 1.$$

Put $d = m + 2 - 2h - 2p$ and $\gamma = M_r r^{-d}$. If $\gamma\|D\| < 1$, the nonlinear remainder in the corner after its first variation obeys

$$\left| [t^k]e_0^* \left((H + D - tU)^{-1} - (H - tU)^{-1} + (H - tU)^{-1}D(H - tU)^{-1} \right) e_m \right| \leq R^{-k} \frac{M_0^2 \gamma \|D\|^2}{1 - \gamma\|D\|}.$$

Proof. The support statements follow by multiplying the indicated shifts by a matrix of lower bandwidth p . If $W = (H - tU)^{-1}$, the weighted inverse bound gives $\|BWA\| \leq M_r r^{-d}$. Each term beyond the first variation in the resolvent identity contains this block between two copies of D . Summing its geometric series gives the stated uniform bound on the disk; Cauchy's coefficient estimate supplies R^{-k} . The same series proves invertibility of the perturbed pencil on that disk. $\square$

Proposition 2.4 (Geometric factor and exponential separation). *Let c extend holomorphically to $|z| > 1$, continuously to $\mathbb{T}$, and meromorphically to infinity with pole order $b \geq 0$ and nonzero leading coefficient. Suppose c is nonzero on $\mathbb{T}$ with winding zero. There are exactly b zeros $\zeta_1, \dots, \zeta_b$ in $|z| > 1$, counting multiplicity. Put*

$$p(z) = \prod_{j=1}^b (1 - z/\zeta_j), \quad F(z) = \frac{p(z^{-1})}{c(z^{-1})}.$$

The apparent singularity of F at zero is removable, and F is the generating function of the first row of $T(c)^{-1}$.

Fix $1 < r < \min_j |\zeta_j|$ (any $r > 1$ in the zero-free exterior if $b = 0$), and write $F(z) = \sum_{j \geq 0} f_j z^j$. There are constants C, N_0 such that, for $N \geq N_0$ and $1 \leq l \leq N - b$,

$$\left(\sum_{j=0}^{l-1} |(T_N(c)^{-1})_{0j} - f_j|^2 \right)^{1/2} \leq C r^{-(N-b-l+1)}. \quad (2)$$

The constants are uniform over a compact holomorphic family of such symbols when the exterior pole order is fixed, the leading coefficient stays nonzero, and the zeros stay outside the chosen radius r . For a variation $\delta c(z) = z^{-j}$, $j \geq 1$, the coefficients of the outside-zero polynomial satisfy $\|\delta p\| = O(r^{-j})$, with any fixed coefficient norm and the same compact-family qualification.

Proof. Nonvanishing and continuity on the boundary give a zero-free collar. The argument principle on a large annulus gives the zero count b . Choose the nonzero constant G so that $c_+ = Gp$, $c_- = c/(Gp)$ satisfy $c_-(\infty) = 1$. Both factors and their inverses are bounded and holomorphic on their respective sides of the circle. The elementary Toeplitz product identity gives $T(c) = T(c_-)T(c_+)$, and hence $T(c)^{-1} = T(c_+^{-1})T(c_-^{-1})$. Its first row has generating function $G^{-1}c_-^{-1}(z^{-1}) = F(z)$.

The symbols $c(rz)$ are again continuous, nonvanishing and of index zero. Finite-section stability, in the form read in [2, Section 3, equation (6)], gives a uniform bound for

$$D_r T_N(c)^{-1} D_r^{-1} = T_N(c(r \cdot))^{-1}, \quad D_r = \text{diag}(1, r, \dots, r^{N-1}).$$

Truncating the infinite inverse row f leaves a residual supported only in the last b columns: $c_k = 0$ for $k > b$, so a missing index $i \geq N$ can contribute to column j only when $j \geq N - b$. Its norm is bounded by a constant times $\|f\|_2$. Multiplication by the finite inverse and its displayed weighted bound then proves (2). Uniformity over compact families follows by a finite covering and the inverse perturbation identity, applied also to the scaled symbols.

For completeness, the coefficients of p depend holomorphically on a holomorphic parameter even when the outside zeros coincide. Their power sums are contour integrals of $z^k c'(z)/c(z)$ on an annulus whose inner boundary is $|z| = r$ and whose outer boundary encloses all zeros. Differentiating and integrating by parts changes the integrand to $-kz^{k-1} \delta c(z)/c(z)$. It is $O(r^{-j})$ for the indicated variation on the two fixed contours. Newton's identities and the nonzero constant term of the monic zero polynomial give the final claim. $\square$

Lemma 2.5 (The finite corner and a squared outside factor). *Under the hypotheses of Proposition 2.4, write $p(z) = \sum_{j=0}^b p_j z^j$, $F(z) = \sum_{j \geq 0} f_j z^j$, and $V_N = T_N(c)^{-1}$. For every fixed allowable $r > 1$,*

$$(V_N)_{0,N-1} = [z^{N-1}] \frac{p(z^{-1})^2}{c(z^{-1})} + O(r^{-N}). \quad (3)$$

The coefficient on the right is a Laurent coefficient; its negative part is finite. Constants are uniform on the compact families in that proposition. No relative error is asserted.

Proof. For $N > b$, truncating the infinite first inverse row gives the exact identity

$$(V_N)_{0,N-1} = f_{N-1} + \sum_{l=0}^{b-1} f_{N+l} \sum_{k=0}^{b-1-l} c_{k+l+1} (V_N)_{0k}.$$

Indeed the omitted tail leaves a residual only in the last b columns, and persymmetry identifies $(V_N)_{N-1-k,N-1}$ with $(V_N)_{0k}$. Equation (2), with $l = b$, replaces the finitely many entries

$(V_N)_{0k}$ by f_k , at error $O(r^{-N})$; the fixed factors r^{2b-1} are absorbed into the constant. The identity $F(z)c(z^{-1}) = p(z^{-1})$ gives

$$\sum_{k=0}^{b-1-l} f_k c_{k+l+1} = p_{l+1}.$$

The resulting sum $\sum_{j=0}^b p_j f_{N-1+j}$ is exactly the coefficient in (3). The case $b = 0$ is the exact upper-triangular identity. $\square$

Lemma 2.6 (Continuation of the far-zero factors across the boundary). *For $g = g_0 + P$ with $P(-1) = 0$ and $h - s$ sufficiently small in $C^1(\mathbb{T})$, the factors p_w can be chosen holomorphically for all w in a neighborhood of $\{w : w^2 - 1 \in \overline{D}\}$. There is a fixed radius $\rho > 1$ such that they consist precisely of the zeros of c_w outside $|z| = \rho$ there. If the positive degree of g is $p > 0$, each factor has degree $2p$; when g has no positive powers the factor is one. There are polynomials A_t and B_t in s , with coefficients holomorphic in a neighborhood of $\overline{D}$, such that*

$$p_w(z)^2 = A_t(z^2) + wzB_t(z^2), \quad t = w^2 - 1. \quad (4)$$

Proof. The function h is holomorphic outside the circle, and h' extends continuously to it: the additional term $2sg_0P$ has derivative $O(|s+1|^{1/2})$ at -1 . For a sufficiently thin closed exterior annulus, $\|h' - 1\| < 1/3$. Any two points of this annulus can be joined in it by a radial segment and a shorter circular arc of total length at most $(1 + \pi/2)$ times their distance. Integration of $h' - 1$ along that path proves injectivity. Thus h maps the annulus conformally to the region between Γ and an enclosing Jordan curve $h(r\mathbb{T})$, for some $r > 1$.

Let U be the bounded interior of this latter curve. For $t \in U$, the zeros of $h(s) - t$ outside $|s| = r$ have constant count $2p$ when $p > 0$, and zero count when $p = 0$, by the argument principle at infinity. The domain $\{w : w^2 - 1 \in U\}$ is connected, since U is simply connected and contains -1 . On $|z| = \sqrt{r}$, neither c_w nor c_{-w} vanishes there. Their equal outside-zero counts are $2p$: at $w = 0$ this follows from $g(s)^2 = (h(s) + 1)/s$ and the argument principle. Choosing $\rho = \sqrt{r}$, contour power sums therefore give holomorphic normalized factors, including at multiple far zeros. For $t \in D$ they agree with the original factors, because the intervening annulus has no zeros. The parity relation $p_{-w}(z) = p_w(-z)$ shows that the even coefficients in p_w^2 are even in w , while its odd coefficients are odd in w . Division of the latter by w is removable at zero. Descent under $w \mapsto w^2 - 1$ proves (4). $\square$

For the factors in the preceding lemma define R_t, X_t by parity:

$$F_w(z) = \frac{p_w(z^{-1})}{g(z^{-2}) - wz} = X_t(z^2) + wzR_t(z^2), \quad t = w^2 - 1.$$

Parity and the removable value at $w = 0$ make both functions holomorphic in t . Their coefficients are the formal inverse rows used below.

Proposition 2.7 (Polynomial stability up to the squared boundary). *For the fixed background of Lemma 2.6, put $\mathcal{W} = \{w : w^2 - 1 \in \overline{D}\}$. There are C, N_0 and $\rho > 1$ such that, for every $w \in \mathcal{W}$ and $N \geq N_0$, $T_N(c_w)$ is invertible and*

$$\|T_N(c_w)^{-1}\| \leq CN^2, \quad \|D_\rho T_N(c_w)^{-1} D_\rho^{-1}\| \leq C, \quad D_\rho = \text{diag}(1, \rho, \dots, \rho^{N-1}).$$

These include the parameters for which c_w vanishes on $\mathbb{T}$. For the odd pencil $W_m(t) = (G^2 - (1 + t)U)^{-1}$ it follows that, uniformly for $t \in \overline{D}$ and large m ,

$$\|W_m(t)\| \leq C(m+1)^4, \quad \|D_r W_m(t) D_r^{-1}\| \leq C, \quad r = \rho^2 > 1. \quad (5)$$

Proof. The scaled symbols $c_w(\rho \cdot)$ are nonzero and of index zero uniformly on the compact parameter set $\mathcal{W}$, by the collar lemma. Uniform finite-section stability gives the second bound in the first display, and also eventual invertibility. This step applies to the scaled symbols, not to a possibly vanishing symbol on $\mathbb{T}$. If $V_N = T_N(c_w)^{-1}$, in particular

$$|(V_N)_{ij}| \leq C\rho^{j-i}.$$

The far-factor function $F_w(z) = p_w(z^{-1})/c_w(z^{-1})$ is holomorphic for $|z| < 1$, also when $w \in \partial\mathcal{W}$. Cancellation removes the far zeros. Near the circle, rationalization of its denominator expresses it as

$$F_w(z) = \frac{p_w(z^{-1})(g(z^{-2}) + wz)}{z^2(h(z^{-2}) - (w^2 - 1))}.$$

The collar map is quantitatively injective. Since $h(s)$ is outside D when $1 < |s| < r$, this gives $\text{dist}(h(s), \overline{D}) \geq c(|s| - 1)$. The numerator is uniformly bounded there. Hence

$$|F_w(z)| \leq \frac{C}{1 - |z|}, \quad |[z^j]F_w(z)| \leq C(j + 1),$$

uniformly in w ; the second inequality is Cauchy's bound at radius $j/(j + 1)$, with the small values of j treated separately.

The formal identity $F_w(z)c_w(z^{-1}) = p_w(z^{-1})$ still holds at boundary parameters. Its row convolution is finite in each column because c_w has fixed positive degree b . Truncation therefore gives a residual of size at most CN supported in the last b columns. Multiplication by the displayed entry bound for V_N shows

$$|(V_N)_{0j} - [z^j]F_w| \leq CN\rho^{j-N+b}.$$

In particular, the first row has entries bounded by CN and $(V_N)_{00} = F_w(0) + O(N\rho^{-N})$. The continuous function $F_w(0)$ never vanishes: it is the ratio of the leading coefficients of the normalized far factor and c_w . Thus $|(V_N)_{00}|$ is bounded below for large N .

For clarity, the elementary Toeplitz inverse identity needed to pass from a row bound to an operator bound is as follows. Write $x = V_N e_0$, $y = V_N e_{N-1}$, and $x_0 = (V_N)_{00}$. If $L(v)$ and $U(v)$ denote lower and upper triangular Toeplitz matrices with first column or row v , respectively, then

$$V_N = \frac{1}{x_0} [L(x)U(y_{N-1}, \dots, y_0) - L(0, y_0, \dots, y_{N-2})U(0, x_{N-1}, \dots, x_1)].$$

One verification uses the equal first and last principal blocks of the original Toeplitz matrix. Their inverse Schur complements give

$$x_0\{(V_N)_{ij} - (V_N)_{i-1,j-1}\} = x_i y_{N-1-j} - y_{i-1} x_{N-j}, \quad i, j \geq 1;$$

iteration with the first row and column gives the identity. The Schur complements exist since $x_0 \neq 0$ and the matrix is persymmetric. Now $\|x\|_1 \leq C$ by the weighted entry estimate, whereas $\|y\|_1 \leq CN^2$ by persymmetry and the first-row bound. The triangle and convolution inequalities prove $\|V_N\| \leq CN^2$.

Finally use the even-odd block matrix in Proposition 2.1. At $w = 0$ its inverse bounds give $\|G^{-1}\| \leq C(m + 1)^2$. Its upper-left inverse block is $W_m(t)G$, so multiplication by G^{-1} proves the unweighted pencil bound. After scaling by ρ in the original variable, the blocks use $G_r = T_{m+1}(g(r \cdot))$, and the off-diagonal factors are w/ρ . Both that full inverse and G_r^{-1} have uniform bounds. The same identity proves the weighted bound in (5). $\square$

Corollary 2.8 (Exponential nonlinear feedback in exact conformal jets). *Fix the preceding background and let f be a conformal map from the disk to D , with $f(0) = 0$. Fix $0 < \sigma < 1/2$, put $q = \lfloor \sigma m \rfloor$, and let*

$$E = \sum_{j=m-q+1}^m \ell_j U^j, \quad V = \sum_j |\ell_j|.$$

For large m , the difference between the perturbed corner $e_0^((G + E)^2 - (1 + f(u))U)^{-1}e_m$ and its affine approximation in E has every u -Taylor coefficient bounded by*

$$\frac{C(m+1)^8 r^{-d} V^2}{1 - Cr^{-d}V}, \quad d = m + 2 - 2q - 2p,$$

provided $Cr^{-d}V < 1$. Here p is the fixed lower bandwidth of G and $r > 1$ is fixed. In particular $q = \lfloor 3m/8 \rfloor$ permits more unknowns than the $m/4$ target jets while preserving an exponentially small nonlinear remainder.

Proof. For large m , $E^2 = 0$. The support and gap argument in Lemma 2.3 applies with its h replaced by q . Use $\|W_m\| \leq C(m+1)^4$, $\|D_r W_m D_r^{-1}\| \leq C$, and $\|GE + EG\| \leq CV$, uniformly on $\overline{D}$. This gives the displayed supremum bound for the nonlinear remainder throughout D . Composition with f and Cauchy's estimate on circles with radii increasing to one give the same bound for every coefficient, without an exponentially growing coefficient-extraction factor. $\square$

Corollary 2.9 (Uniform replacement of the finite Jacobian on the boundary). *Fix $0 < \sigma < 1$ and $q = \lfloor \sigma m \rfloor$ in the preceding setting. For $0 \leq d < q$, let $J_{m,d}(t)$ be the exact corner derivative with respect to $g_{-(m-d)}$. Then*

$$\sup_{t \in \overline{D}} \left| J_{m,d}(t) + 2[s^d]R_t(s)X_t(s) \right| \leq C(m+1)^3 r^{-(m+1-q-2p)}.$$

After composition with a fixed conformal map $f : \mathbb{D} \rightarrow D$, the same bound holds for every Taylor coefficient of the difference. In particular, no exponentially increasing coefficient-extraction factor is needed when replacing the finite Jacobian in conformal jets.

Proof. Decompose the unsquared factor as $p_w(z) = a_t(z^2) + wz b_t(z^2)$. Its coefficient functions, like those of A_t, B_t , are holomorphic through $\overline{D}$. Rationalizing the first-row generating function gives

$$R_t(s) = \frac{\zeta a_t(\zeta) + \zeta^2 b_t(\zeta)g(\zeta)}{h(\zeta) - t}, \quad X_t(s) = \frac{\zeta a_t(\zeta)g(\zeta) + \zeta(1+t)b_t(\zeta)}{h(\zeta) - t}, \quad \zeta = s^{-1}.$$

Both are holomorphic in the disk, including at $s = 0$ after cancellation. The collar distance bound in Proposition 2.7 shows that each has modulus at most $C/(1-|s|)$, uniformly on $\overline{D}$. Their coefficients are consequently bounded by $C(j+1)$.

Let $G_\infty = T(g)$. The formal infinite first-row equations, whose sums in each column are finite because the positive bandwidth is p , give $X_t = R_t G_\infty$ and $R_t(G_\infty^2 - (1+t)U_\infty) = e_0^*$. These identities also hold at $t = -1$ by the displayed analytic formulas. Truncate the row R_t at length $L = m+1$. Its residual for the finite pencil is supported in the last $2p$ columns and is bounded by CL : all omitted indices are at most $L+2p$, and the fixed Wiener norm of g bounds the convolutions. The weighted pencil inverse bound therefore gives

$$|(e_0^* W_m(t))_j - [s^j]R_t(s)| \leq CLr^{j-L+2p}.$$

For $j < q$ multiplication by G gives the same estimate for the finite X row and $[s^j]X_t$, using $q+p < L$ and the finite weighted Wiener norm of $g(r \cdot)$. In the coefficient of degree $d < q$ of their product, summing at most L terms now gives the stated $CL^3 r^{q-L+2p}$ bound. The exact derivative identity (1) supplies the factor -2 . Composition with f gives a bounded analytic difference on the unit disk; Cauchy's estimate with radii tending to one proves the final claim. $\square$

Proposition 2.10 (Exact limiting Jacobian as a boundary integral). *For a polynomial $V(s) = \sum_{d \geq 0} v_d s^d$, define*

$$\mathcal{J}V(t) = -2 \sum_{d \geq 0} v_d [s^d] R_t(s) X_t(s), \quad t \in D.$$

Put

$$N_t(\zeta) = \zeta \left(A_t(\zeta) g(\zeta) + \frac{1}{2} B_t(\zeta) (h(\zeta) + t + 2) \right).$$

Then

$$\mathcal{J}V(t) = -\frac{2}{2\pi i} \int_{\mathbb{T}} \frac{N_t(\zeta) V(\zeta)}{(h(\zeta) - t)^2} d\zeta. \quad (6)$$

Let $\mathcal{C}_\Gamma f(t) = (2\pi i)^{-1} \int_\Gamma f(x)/(x-t) dx$, and set

$$q(\zeta) = \frac{N_{h(\zeta)}(\zeta)}{h'(\zeta)}, \quad L_t(\zeta) = \frac{N_t(\zeta) - N_{h(\zeta)}(\zeta)}{t - h(\zeta)}.$$

The latter quotient is removable at $t = h(\zeta)$. Formula (6) becomes

$$\mathcal{J}V = -2 \partial_t \mathcal{C}_\Gamma [(qV) \circ h^{-1}] + 2 \mathcal{C}_\Gamma [(L_t V/h') \circ h^{-1}].$$

Its leading weight has exactly the square-root endpoint zero:

$$q(\zeta) = \frac{\zeta g(\zeta)}{h'(\zeta)} p_{\sqrt{\zeta} g(\zeta)} (\sqrt{\zeta})^2, \quad \lim_{\zeta \rightarrow -1, \zeta \in \mathbb{T}} \frac{q(\zeta)}{g_0(\zeta)} = -p_0(i)^2 \neq 0.$$

The expression is independent of the choice of $\sqrt{\zeta}$.

Proof. The odd part of $F_w(z)^2$ is $2wzR_t(z^2)X_t(z^2)$. Use (4) in F_w^2 , put $s = z^2$ and $\zeta = s^{-1}$, and combine the two denominators. Direct algebra gives

$$R_t(\zeta^{-1})X_t(\zeta^{-1}) = \frac{\zeta^2 \{A_t(\zeta)g(\zeta) + \frac{1}{2}B_t(\zeta)(h(\zeta) + t + 2)\}}{(h(\zeta) - t)^2}.$$

Coefficient extraction proves the integral identity, first on a nearby exterior circle and then on $\mathbb{T}$ by continuity. Splitting $N_t = N_h + (t-h)L_t$ gives the two Cauchy terms with the displayed signs. At $t = h(\zeta)$ take $w = \sqrt{\zeta} g(\zeta)$ in (4); its right side is $A_h + \zeta g B_h$. This proves the formula for q . Finally $g/g_0 \rightarrow 1$, $h' \rightarrow 1$ and p_0 is even with all its zeros outside $\rho > 1$. These facts prove the endpoint limit and its nonvanishing. $\square$

3 Correction norms and endpoint restoration

The base triangular Jacobian, indexed by reversed correction degree, and its polynomial inverse are

$$(J^0 v)_k = -2(k+1) \sum_{d \geq k} \binom{1/2}{d-k} v_d,$$

$$v_d = -\frac{1}{2} \sum_{k=d}^{q-1} \binom{-1/2}{k-d} \frac{x_k}{k+1}, \quad x = J^0 v. \quad (7)$$

Lemma 3.1 (Localized invisible restoration and the squared curve). *Let $m \geq 2$, let $L(s) = \sum_{j=1}^m \ell_j s^{-j}$, and put*

$$V = \sum_j |\ell_j|, \quad T = \sum_{j \in \mathbb{Z}} |d_j - d_{j-1}|, \quad d_j = j(-1)^j \ell_j,$$

where coefficients outside $1 \leq j \leq m$ are zero. Define

$$K_m(s) = (-s)^{-m-1} \left(\frac{1-s^{-1}}{2} \right)^m, \quad D(s) = L(s) - L(-1)K_m(s).$$

Then $D(-1) = 0$. The added restoring packet has coefficient ℓ^1 norm $|L(-1)| \leq V$, and its frequencies in $zD(z^2)$ are invisible to T_{2m+1} . For $g_0(s) = \sqrt{1+s^{-1}}$,

$$\|g_0 D\|_{C^1(\mathbb{T})} \leq C \left(m^{3/4} V + \sqrt{mVT} \right). \quad (8)$$

If P is fixed and C^1 with $P(-1) = 0$, then

$$\|s(g_0 + P + D)^2 - s(g_0 + P)^2\|_{C^1} \leq C_P \left(m^{3/4} V + \sqrt{mVT} + T + \sqrt{m}V + mV^2 \right).$$

All C^1 norms refer to the angular variable.

Proof. The binomial expansion has alternating coefficients of total absolute sum one. Its support is $-m-1, \dots, -2m-1$ in s , proving the cost and invisibility statements. Write $s = -e^{i\theta}$ and let d be the distance of θ to $2\pi\mathbb{Z}$. The square-root factor satisfies $|g_0| \leq C\sqrt{d}$ and $|\partial_\theta g_0| \leq C/\sqrt{d}$. Summation by parts and the trivial derivative bound give

$$|\partial_\theta L| \leq C \min(mV, T/d), \quad |L(s) - L(-1)| \leq CV \min(md, 1).$$

The first bound follows by multiplying the derivative Fourier series by $1 - e^{-i\theta}$ and taking the absolute coefficient sum.

The packet has $|K_m| = |\cos(\theta/2)|^m$. Differentiating its explicit formula, and using the Gaussian bound near zero and exponential decay away from zero, gives

$$\sup \sqrt{d} |\partial_\theta K_m| \leq Cm^{3/4}, \quad \sup d |\partial_\theta K_m| \leq C\sqrt{m}, \quad |1 - K_m| \leq C \min(md, 1).$$

Hence $|D| \leq CV \min(md, 1)$, so $\|g'_0 D\|_\infty \leq C\sqrt{m}V$. Also

$$\sqrt{d} |L'| \leq C \min(mV\sqrt{d}, T/\sqrt{d}) \leq C\sqrt{mVT}.$$

These estimates prove (8). At the endpoint the product is continuously differentiable, with derivative zero, because $D(-1) = 0$ and D is differentiable.

Since $|P(s)| \leq C_P d$, the same bounds give $\|PD\|_{C^1} \leq C_P(V + T + \sqrt{m}V)$. Finally $\|D\|_\infty \leq 2V$ and $\|D'\|_\infty \leq CmV$, so $\|D^2\|_{C^1} \leq CmV^2$. Expanding the two squares proves the claim. $\square$

Lemma 3.2 (Coefficient and geometry bounds from the base preconditioner). *Let $q \leq m/2$, let x have degree less than q , and define v by (7). Put*

$$L(s) = s^{-m} \sum_{d=0}^{q-1} v_d s^d, \quad D(s) = L(s) - L(-1)K_m(s), \quad R_1 = \sum_{k < q} \frac{|x_k|}{k+1}, \quad R_{1/2} = \sum_{k < q} \frac{|x_k|}{\sqrt{k+1}},$$

where K_m is the localized invisible packet from Lemma 3.1. Then

$$\|L\|_{\ell^1} \leq CR_{1/2}, \quad \text{Var}\{(-1)^d v_d\}_{d \in \mathbb{Z}} \leq R_1, \quad T_m \leq C(mR_1 + R_{1/2}).$$

For $\gamma \geq 0$ write $\|a\|_{\mathcal{W}^\gamma} = \sum_j (1 + |j|)^\gamma |a_j|$. For every $0 \leq \alpha < 1/2$,

$$\|g_0 D\|_{\mathcal{W}^{1+\alpha}} \leq C_\alpha (m^{1+\alpha} R_1 + m^{3/4+\alpha} R_{1/2}). \quad (9)$$

If P is any fixed Laurent polynomial with $P(-1) = 0$, then

$$\|s(g_0 + P + D)^2 - s(g_0 + P)^2\|_{\mathcal{W}^{1+\alpha}} \leq C_{P,\alpha} (m^{1+\alpha} R_1 + m^{3/4+\alpha} R_{1/2} + m^{1+\alpha} R_{1/2}^2).$$

Proof. Set $A_l = \binom{2l}{l}/4^l$. For one input coefficient x_k , the modulated v sequence is a multiple of the reversed finite sequence $A_0, \dots, A_k$. Its zero-padded variation is twice its maximum. The factor $1/(2(k+1))$ in (7) therefore gives the stated variation bound. The identity $\sum_{l=0}^k A_l = (2k+1)A_k = O(\sqrt{k+1})$ gives the coefficient norm bound. Multiplication by the physical frequency $m-d$ gives the bound for T_m .

A useful exact identity is, for $j \geq 1$,

$$\sum_{d=0}^k \binom{1/2}{d+j} \binom{-1/2}{k-d} = \frac{k+1/2}{k+j} \binom{-1/2}{k} \binom{-1/2}{j-1}.$$

It follows by differentiating $(1+z)^{1/2} \sum_{l=0}^k \binom{-1/2}{l} z^l$. The absolute values of these negative Laurent coefficients sum to one: their alternating signs are all opposite to the surviving nonnegative coefficient when evaluated at $s = -1$. Equivalently, apply monotone convergence to the corresponding positive series as its argument increases to one. Including the factor $1/(2(k+1))$ shows that the full Wiener norm of g_0 times this basis correction is $1/(k+1)$. Summing gives

$$\|g_0 L\|_{\mathcal{W}^0} \leq R_1.$$

To estimate the restoring packet, modulate the variable to $u = -s^{-1}$. Its binomial factor is $b_m(u) = ((1+u)/2)^m$. Unimodality gives $\|(1-u)b_m\|_{\ell^1} \leq Cm^{-1/2}$. Write $(1-u)^{1/2} = 1 - \sum_{l \geq 1} c_l u^l$, where $c_l \geq 0$, $\sum c_l = 1$ and $c_l \leq Cl^{-3/2}$. Then

$$\|(1-u)^{1/2} b_m\|_{\ell^1} \leq \sum_{l \geq 1} c_l \min(2, Clm^{-1/2}) \leq Cm^{-1/4}.$$

On frequencies of magnitude at most $4m+4$, these two Wiener bounds, $|L(-1)| \leq CR_{1/2}$, and multiplication by the largest weight prove (9).

For completeness, the infinite tail is controlled by the endpoint cancellation, not by a divergent first moment of g_0 . In the modulated variable D is a polynomial supported at degrees at most $2m+1$, with coefficient sum zero and coefficient norm at most $CR_{1/2}$. For $n > 4m+4$ the difference of two square-root kernel coefficients whose indices differ by $j \leq 2m+1$ is bounded by $Cjn^{-5/2}$, by the binomial recurrence. Thus the product's n th coefficient is at most $CmR_{1/2}n^{-5/2}$. Its weighted tail is bounded by $C_\alpha m^{1/2+\alpha} R_{1/2}$, which is smaller than the restoring term already retained. This proves the first weighted estimate.

Finally factor $P = (1+s)Q$ with Q a fixed Laurent polynomial. The variation bound gives $\|(1+s)L\|_{\ell^1} \leq R_1$. The binomial-packet difference bound gives $\|(1+s)K_m\|_{\ell^1} \leq Cm^{-1/2}$. These are finite polynomials of frequency $O(m)$, so their $\mathcal{W}^{1+\alpha}$ norms give the required bound for PD . The term D^2 has coefficient norm at most $CR_{1/2}^2$ and frequency $O(m)$. Expanding the squared curve proves the last assertion. $\square$

Corollary 3.3 (A sufficient negative-Sobolev correction size). *Fix $-1/2 < s < 0$ and $q \leq m/2$. If $\|x\|_{H^s} \leq C_P m^{s-1}$, the preceding correction satisfies*

$$\|L\|_{\ell^1} = O_P(m^{-1}), \quad T_m = O_P(m^s).$$

For every $0 \leq \alpha < \min(-s, 1/4)$ its restored squared-curve change tends to zero in $\mathcal{W}^{1+\alpha}$, with bound

$$C_{P,s,\alpha}(m^{s+\alpha} + m^{\alpha-1/4} + m^{\alpha-1}).$$

Consequently the construction can preserve a small $C^{1,\alpha}$ neighborhood, not just a C^1 neighborhood, if this norm estimate for the preconditioned correction is established.

Proof. Weighted Cauchy–Schwarz gives $R_1 \leq C_s \|x\|_{H^s}$ and $R_{1/2} \leq C_s q^{-s} \|x\|_{H^s}$. The former uses $s > -1/2$, the latter $s < 0$. Since $q \leq m/2$, these give $R_1 = O_P(m^{s-1})$ and $R_{1/2} = O_P(m^{-1})$. Substitute in the preceding lemma. Absolute weighted Fourier summability implies the stated $C^{1,\alpha}$ control. $\square$

Lemma 3.4 (The restored perturbation divided by the endpoint square root). *For the localized restoration of any negative correction L supported in frequencies of magnitude at most m , put $V = \|L\|_{\ell^1}$. For every $0 \leq r < 1$, in the two-sided circle Sobolev norm,*

$$\|D/g_0\|_{H^r(\mathbb{T})} \leq C_r m^r V \sqrt{\log(m+1)}.$$

In particular, corrections with $V = O_P(m^{-1})$ change P/g_0 by an arbitrarily small amount in $H^r(\mathbb{T})$ as $m \rightarrow \infty$. For $1/2 < r < 1$ this also controls the supremum norm.

Proof. The restored polynomial has $\|D\|_{\ell^1} \leq 2V$, frequencies of magnitude at most $2m+1$, and $D(-1) = 0$. At angular distance d from -1 , these facts give $|D| \leq CV \min(md, 1)$, while $|g_0|^2$ is comparable to d . Integration therefore gives $\|D/g_0\|_2 \leq CV \sqrt{\log(m+1)}$. After modulation to $u = -s^{-1}$, the coefficients of $1/g_0$ are $A_n = \binom{2n}{n}/4^n$. Their first differences are $O(n^{-3/2})$. The zero sum of the modulated coefficients of D thus bounds the n th product coefficient by $CmVn^{-3/2}$ for $n > 4m+4$. The low-frequency part of the H^r norm is at most Cm^r times the L^2 norm. The squared high-frequency part is bounded by $Cm^2V^2 \sum_{n>4m} n^{2r-3} \leq C_r m^{2r}V^2$ for $r < 1$. These estimates prove the assertion. The last statement is the coefficient Cauchy–Schwarz embedding for $r > 1/2$. $\square$

4 The geometric operator estimate

The following estimates control the actual Jacobian, including its exterior-zero factors, uniformly as the degree of the background grows.

Lemma 4.1 (The square-root extension of the analytic projection). *Let $g_0(s) = \sqrt{1+s^{-1}}$ and let y be a polynomial. Define $V = T(g_0)^{-1}y$, using the triangular polynomial inverse, and $Ey = g_0V$. For every $0 < r < 1$,*

$$\|Ey\|_{H^r(\mathbb{T})} \leq C_r \|y\|_{H^r}, \quad \Pi_+ Ey = y.$$

For $r > 1/2$, the same inverse also satisfies $\|V\|_{L^2} \leq C_r \|y\|_{H^r}$. These operators therefore extend by density to the indicated spaces. For $r > 1/2$ the continuous representative of Ey vanishes at -1 .

Proof. Write $A_l = \binom{2l}{l}/4^l$. The exact partial-convolution identity in Lemma 3.2 gives, up to signs, the matrix of the negative part of E :

$$K_{j,k} = \frac{(k+1/2)A_k A_{j-1}}{k+j}, \quad j \geq 1, \quad k \geq 0.$$

Since $A_l \leq C(l+1)^{-1/2}$, the matrix after the Sobolev weights is bounded entrywise by

$$C \frac{j^{r-1/2}(k+1)^{1/2-r}}{j+k}.$$

Apply Schur's test with row weights $j^{-1/2}$ and column weights $(k+1)^{-1/2}$. Its two required sums are bounded by

$$Cj^r \sum_{k \geq 0} \frac{(k+1)^{-r}}{j+k}, \quad C(k+1)^{1-r} \sum_{j \geq 1} \frac{j^{r-1}}{j+k},$$

respectively. Splitting each sum at $j = k+1$, or comparing with the corresponding integral, bounds both uniformly for $0 < r < 1$. This proves the first assertion.

The coefficient matrix of $T(g_0)^{-1} : H^r \rightarrow L^2$ has squared Hilbert–Schmidt norm

$$\sum_{k \geq 0} (k+1)^{-2r} \sum_{d=0}^k A_{k-d}^2 \leq C \sum_{k \geq 0} (k+1)^{-2r} \log(k+2) < \infty$$

when $r > 1/2$. For polynomial inputs $Ey = g_0V$ vanishes at -1 . Point evaluation is continuous on $H^r(\mathbb{T})$ for $r > 1/2$, so the last assertion follows by density. $\square$

Lemma 4.2 (Small Cauchy-projection perturbations in the circle parameter). *Let $h(s) = s + k(s)$, with $k \in \mathcal{W}^{1+\alpha}$ for some $\alpha > 0$, and put $\delta = \sum_n |n| |k_n|$. If δ is sufficiently small, $h(\mathbb{T})$ is a positively oriented Jordan curve. Let C_h denote the interior Cauchy projection pulled back by h . For $0 \leq r \leq 1$,*

$$\|C_h - \Pi_+\|_{H^r(\mathbb{T}) \rightarrow H^r(\mathbb{T})} \leq C_r \delta.$$

Write $k = k_- + k_+$, where k_- has nonpositive powers and k_+ has strictly positive powers. If, in addition, $\|k_+\|_{\mathcal{W}^{r+1}}$ is sufficiently small, then

$$\|\Pi_+ C_h \Pi_-\|_{L^2 \rightarrow H^r} \leq C_r \|k_+\|_{\mathcal{W}^{r+1}}.$$

Here $\Pi_- = I - \Pi_+$; the constants do not involve higher derivatives of k_- .

Proof. The divided difference

$$H(\zeta, \eta) = \frac{h(\zeta) - h(\eta)}{\zeta - \eta}$$

belongs to the two-variable Wiener algebra and $\|H - 1\|_{\mathcal{W}(\mathbb{T}^2)} \leq \delta$. Indeed the divided difference of a monomial of degree n has $|n|$ terms of coefficient modulus one, for either sign of n . Thus $|h(\zeta) - h(\eta)| \geq (1 - \delta)|\zeta - \eta|$. The same statement along the homotopy $s + tk(s)$ proves the orientation assertion. The constant term of k causes only a translation.

Set $A(\zeta, \eta) = h'(\zeta)/H(\zeta, \eta)$, where h' is the derivative with respect to the circle variable. The Wiener Neumann series gives

$$\|A - 1\|_{\mathcal{W}(\mathbb{T}^2)} \leq \frac{2\delta}{1 - \delta}.$$

If $A = \sum a_{j,l} \zeta^j \eta^l$, the Cauchy boundary formula is the norm-convergent operator series

$$C_h = \sum_{j,l} a_{j,l} M_{s^l} \Pi_+ M_{s^j}.$$

The half-jump terms agree because $A(\eta, \eta) = 1$. This identity follows first with radial regularization of the circle Cauchy kernel and then by its L^2 limit; the $C^{1,\alpha}$ boundary has the usual Cauchy boundary values. It proves the asserted small bound on L^2 . The operator is a projection. This follows from the Cauchy formula for smooth curves and then from Fourier approximation of h : the displayed Wiener representation gives operator-norm convergence on L^2 , so the identity $C_h^2 = C_h$ passes to the limit.

For smooth v , integration by parts in the Cauchy integral gives

$$\partial_\theta(C_h v) = h_\theta C_h(v/h_\theta).$$

The multipliers h_θ and h_θ^{-1} are uniformly bounded on L^2 and are within $C\delta$ there of is and $(is)^{-1}$. Subtracting the analogous identity for Π_+ and using the L^2 estimate therefore gives the small H^1 bound. The formula and bound for the stated regularity follow by Fourier approximation of h and density of smooth v . Interpolation between L^2 and H^1 proves the estimate for $0 < r < 1$.

For the last claim use the two-variable algebra with coefficient weight $(1 + \max(j, 0) + \max(l, 0))^r$. The divided differences of k_- have only nonpositive powers in each variable, and their weighted norm is at most δ . Those of k_+ have weighted norm at most $C_r \|k_+\|_{\mathcal{W}^{r+1}}$. Neumann expansion consequently splits $A = A_- + A_{\text{rem}}$, where A_- has only nonpositive powers in both variables and

$$\|A_{\text{rem}}\|_{\mathcal{W}_{\text{pos}}^r(\mathbb{T}^2)} \leq C_r \|k_+\|_{\mathcal{W}^{r+1}}.$$

Every term of $\Pi_+ C_h \Pi_-$ coming from A_- vanishes. A term $\Pi_+ M_{s^l} \Pi_+ M_{s^j} \Pi_-$ can be nonzero only for $j \geq 1$ and $j + l \geq 1$. It is a partial shift from negative input frequencies to output frequencies between 0 and $j + l - 1$. Its $L^2 \rightarrow H^r$ norm is therefore at most $(1 + \max(j, 0) + \max(l, 0))^r$. Summing the remainder coefficients proves the final estimate. $\square$

Lemma 4.3 (An endpoint Bezout kernel in the Wiener algebra). *Let $b(x) = \sqrt{1+x}$ and let D be a polynomial with $D(-1) = 0$. For every $\alpha > 0$,*

$$\left\| \frac{b(y)D(x) - b(x)D(y)}{y - x} \right\|_{\mathcal{W}(\mathbb{T}^2)} \leq C_\alpha \|bD\|_{\mathcal{W}^{1+\alpha}}.$$

The assertion is used only when its right side is finite.

Proof. Put $v = bD$ and $q = v/(1+x)$. The kernel on the left is $b(x)b(y)(q(x) - q(y))/(y - x)$. Since $v(-1) = 0$, write

$$v(x) = \sum_{n \geq 1} v_n (x^n - (-1)^n).$$

For one summand $x^n - (-1)^n$, its quotient by $1 + x$ is an alternating geometric polynomial. After replacing x, y by $-x, -y$, its kernel is, up to sign,

$$\sqrt{1-x}\sqrt{1-y} T_{n-2}(x, y), \quad T_N = \sum_{\substack{i,j \geq 0 \\ i+j \leq N}} x^i y^j.$$

The case $n = 1$ has zero kernel.

Write $\sqrt{1-x} = \sum_{\ell \geq 1} c_\ell (1-x^\ell)$, where $c_\ell > 0$, $\sum c_\ell = 1$, and $\sum_{\ell \geq t} c_\ell = \binom{2t-2}{t-1}/4^{t-1} \leq C/\sqrt{t}$. Regard the coefficients of T_{n-2} as a triangular indicator on $\mathbb{Z}^2$. A horizontal translation by ℓ changes its ℓ^1 norm by at most $Cn \min(\ell, n)$, and the analogous vertical bound holds. Consequently

$$\|(I - S_x^\ell)(I - S_y^k)T_{n-2}\|_{\ell^1} \leq Cn \min(\ell, k, n).$$

Summing against $c_\ell c_k$ gives

$$\|\sqrt{1-x}\sqrt{1-y}T_{n-2}\|_{\mathcal{W}} \leq Cn \sum_{t=1}^n \left(\sum_{\ell \geq t} c_\ell \right)^2 \leq Cn \log(n+1).$$

Finally $n \log(n+1) \leq C_\alpha (1+n)^{1+\alpha}$. The resulting absolutely convergent kernel series agrees with the displayed divided difference in the open bidisk and hence on its continuous boundary extension. $\square$

Lemma 4.4 (The unsquared divided difference). *Write $g = g_0 + P_- + P_+$, where P_- has strictly negative powers, P_+ has strictly positive powers, and both vanish at -1 . For $a(z) = zg(z^2)$ and $a_0(z) = zg_0(z^2)$, set*

$$F(z, \eta) = \frac{a(z) - a(\eta)}{a_0(z) - a_0(\eta)} = F_-(z, \eta) + F_+(z, \eta),$$

where $F_- - 1$ and F_+ are the contributions of P_- and P_+ . All quotients have continuous extensions at their removable boundary points. For $\alpha > 0$,

$$\|F_- - 1\|_{\mathcal{W}(\mathbb{T}^2)} \leq C_\alpha \|g_0 P_-\|_{\mathcal{W}^{1+\alpha}},$$

and $F_- - 1$ has strictly negative powers of each variable. For $K \geq 0$, let $\mathcal{W}_{\text{pos}, z}^K$ have coefficient weight $(1 + \max(j, 0))^K$ on the exponent j of z , with no weight on the exponent of η . Then

$$\|F_+\|_{\mathcal{W}_{\text{pos}, z}^K} \leq C_K \|P_+\|_{\mathcal{W}^{K+2}}.$$

Proof. Rationalizing the negative contribution separates it into

$$\frac{sg_0(s)P_-(s) - tg_0(t)P_-(t)}{s-t} + z\eta \frac{g_0(t)P_-(s) - g_0(s)P_-(t)}{s-t}, \quad s = z^2, \quad t = \eta^2.$$

The first divided difference has Wiener norm bounded by $C\|g_0 P_-\|_{\mathcal{W}^1}$. For the second put $x = s^{-1}$, $y = t^{-1}$ and $D(x) = P_-(x^{-1})$. It becomes $z^{-1}\eta^{-1}$ times the kernel in Lemma 4.3. The claimed norm and Fourier orientation follow. In particular the quotient extends at the two distinct contact parameters where both base values are zero.

For the positive contribution write $P_+(s) = (1+s)Q(s)$ and $r(z) = zQ(z^2)$, an odd polynomial. Put $A = a_0(z)$ and $B = a_0(\eta)$. Since the change in a is $A^2 r(z)$, exact algebra gives

$$F_+ = Ar(z) + Br(\eta) + AB \frac{A+B}{z+\eta} \frac{r(z) - r(\eta)}{z-\eta}.$$

The two-variable kernel $AB(A+B)/(z+\eta)$ has finite Wiener norm and has at most one positive power in each variable. Indeed, with $x = z^{-1}$, $y = \eta^{-1}$, $b_x = \sqrt{1+x^2}$ and $b_y = \sqrt{1+y^2}$, it equals

$$(xy)^{-1} \left((1+x^2)b_y + x(y-x) \frac{b_x b_y}{b_x + b_y} \right).$$

The last fraction belongs to the bidisk Wiener algebra because

$$\frac{b_x b_y}{b_x + b_y} = \int_0^\infty b_x e^{-ub_x} b_y e^{-ub_y} du$$

is a convergent Bochner integral there. To verify convergence, the elementary subordination formula

$$e^{-u\sqrt{1+x^2}} = \frac{u}{2\sqrt{\pi}} \int_0^\infty v^{-3/2} e^{-u^2/(4v)} e^{-v(1+x^2)} dv$$

gives Wiener norm at most one. Its positive scalar density has mass one, and its u derivative has total variation C/u , by the substitution $v = u^2 w$. Thus $\|b_x e^{-ub_x}\|_{\mathcal{W}} \leq C/u$; the bound 2 for small u follows instead from $\|b_x\|_{\mathcal{W}} \leq 2$. The product is integrable in u . The formula is first checked in the open bidisk, where the square roots have positive real part, and then extended by the Wiener convergence.

A monomial of degree n in r now contributes at most $C_K(1+n)^{K+1}$ to the required norm: its divided difference has n terms, and the resulting positive z degree is at most $n+1$. Finally division by $1+s$, using $P_+(-1) = 0$, gives $\|Q\|_{\mathcal{W}^{K+1}} \leq C_K \|P_+\|_{\mathcal{W}^{K+2}}$. This proves the last assertion. $\square$

Lemma 4.5 (A quantitative conformal coordinate near the circle). *For fixed $0 < \alpha < 1$, a sufficiently small $\mathcal{W}^{1+\alpha}$ neighborhood of $h(s) = s$ admits normalized Riemann maps $f : \mathbb{D} \rightarrow D$ with*

$$\|f' - 1\|_\infty \leq C_\alpha \|h - s\|_{\mathcal{W}^{1+\alpha}}.$$

The maps have uniformly bounded $C^{1,\alpha}$ norms on the closed disk. The smallness threshold and bound follow from elementary Hölder norm estimates and can be made effective. For the real-symmetric curves used here, $f(-1) = -1$.

Proof. Smallness in the stated Fourier norm implies smallness in $C^{1,\alpha}(\mathbb{T})$. The angle of $h(e^{it})$ is a $C^{1,\alpha}$ diffeomorphism close to t . Inverting it represents the curve as $\rho(\theta)e^{i\theta}$, with

$$\|\ell\|_{C^{1,\alpha}} \leq C_\alpha \|h - s\|_{\mathcal{W}^{1+\alpha}}, \quad \ell = \log \rho.$$

This estimate follows directly by differentiating the angle and the logarithm of $h(e^{it})e^{-it}$, and applying the product, composition, and inverse-function estimates in $C^{1,\alpha}$. All denominators stay uniformly away from zero.

Let H be the periodic Hilbert transform, with zero mean and the sign for which it supplies the imaginary part of an analytic function from its real part. It is bounded on C^α . For completeness, subtract the value at the evaluation point in its cotangent-kernel formula; splitting at the distance between two evaluation points bounds the Hölder seminorm by $C(\alpha^{-1} + (1-\alpha)^{-1})$ times the input seminorm. It commutes with differentiation and has the corresponding bound on $C^{1,\alpha}$.

Seek a real mean-zero angle correction u satisfying

$$u = H[\ell(t + u(t))].$$

Use the closed set $\|u\|_{C^{1,\alpha}} \leq M$, where $M = C_\alpha \|\ell\|_{C^{1,\alpha}} < 1/4$, with a sufficiently large fixed coefficient, but give this set the C^α metric. The set is complete: a C^α limit with these uniform derivative bounds is still $C^{1,\alpha}$ with the same bound, by compactness of the derivatives and the fundamental theorem of calculus. The Hilbert-transform and composition estimates show that the displayed map preserves the set for small ℓ . For two inputs use

$$\ell(t + u) - \ell(t + v) = (u - v) \int_0^1 \ell'(t + v + \lambda(u - v)) d\lambda.$$

The intervening angle maps are uniformly Lipschitz. Hölder product and composition bounds therefore give Lipschitz constant $C_\alpha \|\ell\|_{C^{1,\alpha}} < 1$ in the C^α metric. The contraction theorem supplies the required small $C^{1,\alpha}$ solution.

Let G be analytic in the disk with boundary real part $\ell(t + u(t))$, boundary imaginary part $u(t)$, and real $G(0)$. Then $f(z) = z \exp G(z)$ has the prescribed boundary curve, $f(0) = 0$, and $f'(0) > 0$. The norm bounds for G give the displayed derivative estimate; all derivatives extend continuously to the boundary. Poisson extension preserves the C^α class on the closed disk: subtract the value at the nearest boundary point and split the Poisson integral at angular distance $1 - |z|$ to bound the radial error by $C_\alpha(1 - |z|)^\alpha$; angular differences follow by translation. Apply this to the boundary derivative of G . If the neighborhood is small, $\|f' - 1\|_\infty < 1/4$, so integration along disk chords proves univalence and uniform bi-Lipschitz bounds. The boundary homeomorphism and the argument principle identify its image with D . In the symmetric case ℓ is even. Uniqueness makes u odd, hence $u(\pi) = 0$, and $\rho(\pi) = 1$ gives $f(-1) = -1$. $\square$

Proposition 4.6 (Uniform small far factors and spectral derivatives). *Under the hypotheses of Lemma 4.4, suppose that $\|g_0 P_-\|_{\mathcal{W}^{1+\alpha}}$ and $\|P_+\|_{\mathcal{W}^{K+3}}$ are sufficiently small. Assume also the preceding squared-curve neighborhood, so that the normalized f and its inverse have uniform bi-Lipschitz bounds. For the actual far factors*

$$p_w(z)^2 = A_t(z^2) + wzB_t(z^2), \quad t = w^2 - 1,$$

one has, for $K \geq 0$,

$$\sum_{j \geq 0} (1+j)^K (\|[s^j](A_t(s) - 1)\|_{H^\infty(D)} + \|[s^j]B_t(s)\|_{H^\infty(D)}) \leq C_K \|P_+\|_{\mathcal{W}^{K+3}}.$$

The constants are uniform on a sufficiently small fixed neighborhood.

Proof. For a boundary value $w = a(\eta)$ the positive Wiener factor is

$$\log p_{a(\eta)}(z) = \Pi_{>0,z} \log F(z, \eta).$$

This can be checked on a sufficiently thin exterior circle: $(a_0(z) - a_0(\eta))/z$ has a zero-free exterior logarithm with no positive Laurent part, and the positive part of the remaining logarithm is the logarithm of the normalized far factor. The previously proved collar separates its zeros from that circle. Taking its boundary limit gives the displayed identity in the two-variable Wiener algebra.

The function F_- is close to one and coanalytic in both variables. Its logarithm contributes nothing to $\Pi_{>0,z}$. Neumann series in the submultiplicatively weighted algebra $\mathcal{W}_{\text{pos},z}^K$, followed by exponentiation, give

$$\|p_{a(\eta)}(z) - 1\|_{\mathcal{W}_{\text{pos},z}^K} \leq C_K \|P_+\|_{\mathcal{W}^{K+2}}.$$

In particular this bounds the sum of the weighted suprema of the individual positive- z coefficients on the spectral boundary. Their holomorphic continuation in w , established in the collar lemma, and the maximum principle give the same bound on the two closed lobes $\{w : w^2 - 1 \in \overline{D}\}$.

For the odd coefficients we also need a spectral derivative bound. Differentiating the factor on an exterior circle gives

$$\partial_w \log p_w(z) = -\Pi_{>0,z} \frac{1}{a(z) - w}.$$

At $w = a(\eta)$ the reciprocal is

$$\frac{1}{a_0(z) - a_0(\eta)} F(z, \eta)^{-1}.$$

The first factor has only negative powers of z ; every one of its coefficients has Wiener norm at most 2 as a function of η . To see this, expand $(a_0(z) + a_0(\eta))/(z^2 - \eta^2)$ geometrically in z^{-2} . The odd coefficients are partial sums of square-root binomial coefficients times powers of η , and the even coefficients are $a_0(\eta)$ times a power of η .

Subtract F_-^{-1} from F^{-1} : its product with the base reciprocal is coanalytic and disappears under the positive projection. For any remaining series E the coefficient count just described gives

$$\left\| \Pi_{>0,z} \frac{E(z,\eta)}{a_0(z) - a_0(\eta)} \right\|_{\mathcal{W}_{\text{pos},z}^K} \leq C_K \|E\|_{\mathcal{W}_{\text{pos},z}^{K+1}}.$$

Only positive input degrees j contribute, and at most j base coefficients can contribute for each such degree. The series is absolutely convergent in the indicated norm. For the fixed Laurent-polynomial background it also agrees with the exterior-circle projection: the positive Laurent coefficients decay exponentially on its fixed collar, so the radial limit can be taken term by term. Apply the estimate to $E = F^{-1} - F_-^{-1}$ and use the preceding Neumann bounds. It follows that

$$\|\partial_w p_{a(\eta)}(z)\|_{\mathcal{W}_{\text{pos},z}^K} \leq C_K \|P_+\|_{\mathcal{W}^{K+3}}.$$

The same holds for $\partial_w(p_w^2)$, and the maximum principle again bounds the sum of coefficient suprema throughout both lobes.

For an odd coefficient $Q_j(w) = [z^j]p_w(z)^2$, parity gives $Q_j(0) = 0$. There is a path in the corresponding closed lobe from 0 to w of length at most $C|w|$. Indeed, if $w^2 - 1 = f(u)$, use the path $w(v) = \sqrt{1 + f(-1 + v(u+1))}$, $0 \leq v \leq 1$, with the continuous choice of sign. Bi-Lipschitz bounds give $|1 + f(-1 + v(u+1))| \asymp v|u+1|$, so integration of its derivative bounds the length by $C\sqrt{|u+1|} \leq C|w|$. Consequently $|Q_j(w)/w| \leq C \sup |Q'_j|$. The even coefficient bound supplies $A_t - 1$, and the odd coefficient bound just proved supplies B_t . Their weighted sums are exactly the asserted estimate. $\square$

Proposition 4.7 (The two-level isomorphism estimate). *Fix $\alpha = 1/8$ and split the finite real Laurent background as $P = P_- + P_+$, with strictly negative and strictly positive powers, respectively, and with $P_-(-1) = P_+(-1) = 0$. There is a sufficiently small $\gamma > 0$ such that the bounds*

$$\begin{aligned} \|h - s\|_{\mathcal{W}^{1+\alpha}} &< \gamma, & \|P_-\|_{\mathcal{W}^0} &< \gamma, \\ \|P_-/g_0\|_{\mathcal{W}^0} &< \gamma, & \|g_0 P_-\|_{\mathcal{W}^{1+\alpha}} &< \gamma, & \|P_+\|_{\mathcal{W}^4} &< \gamma \end{aligned}$$

imply that the actual operator $\mathcal{A}_g x = (\mathcal{J}_g(J^0)^{-1}x) \circ f$ extends to compatible isomorphisms on $H^{-1/4}$ and $H^{-1/8}$. Their norms and inverse norms are bounded independently of the degree of P .

Proof. The proof applies at each of $r = 3/4, 7/8$, with $s = r - 1$. For a one-variable multiplier use the coefficient norm

$$\|M\|_{\mathcal{W}_{\text{pos}}^r} = \sum_{n \in \mathbb{Z}} (1 + \max(n, 0))^r |M_n|.$$

It is submultiplicative. The endpoint extension lemma gives

$$\|\Pi_+(MEy)\|_{H^r} \leq C_r \|M\|_{\mathcal{W}_{\text{pos}}^r} \|y\|_{H^r}, \quad \|MEy\|_{L^2} \leq C_r \|M\|_{\mathcal{W}^0} \|y\|_{H^r}. \quad (10)$$

Indeed a coanalytic multiplier cannot turn the negative part of Ey positive, and its backward shifts on analytic H^r are contractions. A positive shift by j on two-sided H^r costs at most $C_r(1+j)^r$. This proves both assertions by summation.

Put $R = g/g_0$ and $m = R/h'$. The assumptions imply

$$\|m - 1\|_{\mathcal{W}_{\text{pos}}^r} \leq C_r \gamma, \quad \|(h - s)_+\|_{\mathcal{W}^{r+1}} \leq C_r \gamma.$$

Here are the Fourier details. The negative part $1 + P_-/g_0$ of R is close to one in $\mathcal{W}^0$. Writing $P_+ = (1 + s)Q$ gives $P_+/g_0 = sg_0Q$ and $\|Q\|_{\mathcal{W}^r} \leq C_r \|P_+\|_{\mathcal{W}^{r+1}}$. The part of $h - s$ without positive-background factors, $2sg_0P_- + sP_-^2$, is nonpositive. Every other term is a product containing P_+ , so its positive weighted norm of order $r + 1$ is bounded by $C_r \|P_+\|_{\mathcal{W}^{r+2}}$, using the bounded unweighted norms of the coanalytic factors. Thus $h' - 1$ is small in $\mathcal{W}_{\text{pos}}^r$; Neumann inversion proves the bound for m .

Consider the operator on analytic H^r

$$By = \Pi_+ s^{-1} C_h [smEy].$$

Its direct projection is $\Pi_+ mEy$, within $C_r \gamma$ of y by (10). The positive part of the density $smEy$ is in H^r with bounded norm. Apply the small H^r estimate for $C_h - \Pi_+$ to that part, and the $L^2 \rightarrow H^r$ estimate for $\Pi_+ C_h \Pi_-$ to its negative part. Lemma 4.2 then gives

$$\|B - I\|_{H^r \rightarrow H^r} \leq C_r \gamma.$$

Consequently B is an isomorphism for small γ .

We explain why this is the required bare Cauchy isomorphism, rather than merely a projected estimate. Let $\mathcal{R}_r = C_h H^r(\mathbb{T})$. The small norm of $C_h - \Pi_+$ implies that $\Pi_+ : \mathcal{R}_r \rightarrow H^r$ is an isomorphism: on analytic inputs $\Pi_+ C_h$ is close to the identity, while a range vector v with $\Pi_+ v = 0$ satisfies $v = (C_h - \Pi_+)v$ and is zero. Constants are fixed by C_h . Therefore

$$\pi(v) = \Pi_+ s^{-1} v$$

identifies $\mathcal{R}_r$ modulo constants with analytic H^r . The same range argument on L^2 shows that an L^2 range vector whose positive part is in H^r actually lies in $\mathcal{R}_r$. This applies to the densities just considered.

Composition by $h^{-1} \circ f$ identifies $\mathcal{R}_r$ with analytic H^r , preserving constants, with uniform bounds in both directions. The boundary maps are uniformly bi-Lipschitz. For $0 < r < 1$ their action on $H^r(\mathbb{T})$ follows from the equivalent norm

$$\|v\|_{L^2}^2 + \int_{\mathbb{T}} \int_{\mathbb{T}} \frac{|v(z) - v(\zeta)|^2}{|z - \zeta|^{1+2r}} |dz| |d\zeta|,$$

by change of variables. Equivalence with the Fourier norm follows by expanding in Fourier modes. The range identification then follows from the Cauchy formula, first for analytic polynomials in the disk coordinate and then by density.

Differentiation in the physical spectral variable, followed by composition with f , acts as $Z \mapsto Z'/f'$ in the disk. Differentiation is an isomorphism from analytic H^r modulo constants onto H^{r-1} . Analytic H^∞ multipliers are bounded on H^s for $s < 0$: this follows from the equivalent weighted Bergman norm

$$\int_{\mathbb{D}} |X(u)|^2 (1 - |u|^2)^{-1-2s} dA(u).$$

The equivalence is immediate by integrating each monomial and comparing its beta-integral weight with $(k + 1)^{2s}$. Both f' and $1/f'$ are uniformly bounded here.

Define the bare operator by

$$\mathcal{J}_{\text{bare}}V = -2\partial_t \mathcal{C}_\Gamma[(sgV/h') \circ h^{-1}].$$

Since $y = \Pi_+ g_0 V$ satisfies $x_k = -2(k+1)y_k$ when $x = J^0 V$, the preceding isomorphisms factor $(\mathcal{J}_{\text{bare}}(J^0)^{-1}x) \circ f$ into invertible maps with uniform bounds. This proves the bare isomorphism on H^s . The same proof of boundedness, without requiring a multiplier near one, shows that the differentiated Cauchy transform of a density $sMEy$ has H^s output norm at most $C_r \|M\|_{\mathcal{W}_{\text{pos}}^r} \|y\|_{H^r}$.

It remains to retain the actual exterior factors. Write $A_t(s) = \sum A_j(t)s^j$ and $B_t(s) = \sum B_j(t)s^j$ in Proposition 2.10. The A part of the actual Jacobian is $\sum_j A_j(t)\mathcal{J}_{\text{bare}}(s^j V)(t)$. The shift costs at most $C_r(1+j)^r$ in the preceding density bound. Its coefficient is an analytic multiplier on H^s after composition with f , so it costs only its $H^\infty(D)$ norm.

For the B part use the exact identities $h+t+2 = 2(h+1) - (h-t)$ and $h+1 = sg^2$. The first resulting term is the differentiated Cauchy transform with density $s^{j+2}g^2V/h' = s^{j+2}gmEy$. The general density bound applies with $M = s^{j+1}gm$, at cost at most $C_r(1+j)^r$. The second term is the undifferentiated Cauchy transform with density $s^{j+1}V/h'$. It is bounded from $y \in H^r$ to L^2 , by the second assertion of Lemma 4.1, boundedness of C_h on L^2 , and the bounded multiplier $1/h'$. The boundary composition is bounded on L^2 , which embeds in H^s . This bound does not grow with j .

Proposition 4.6, used with $K = r \leq 1$, now proves

$$\|\mathcal{A}_g - \mathcal{A}_{\text{bare},g}\|_{H^s \rightarrow H^s} \leq C_r \|P_+\|_{\mathcal{W}^4} \leq C_r \gamma.$$

Neumann inversion, after decreasing the single fixed γ , gives the asserted actual isomorphisms for both chosen exponents. All formulas agree on polynomial inputs and extend compatibly. The inverse on the stronger space agrees with the weaker inverse by uniqueness in the weaker space. $\square$

5 The forcing and growing finite systems

Proposition 5.1 (The scattering generating function). *With $\zeta = s^{-1}$ set*

$$S_t(s) = \frac{\zeta A_t(\zeta) + \zeta^2 B_t(\zeta)g(\zeta)}{h(\zeta) - t}.$$

The odd part of $p_w(z^{-1})^2/(g(z^{-2})-wz)$ is $wzS_t(z^2)$. Consequently the corner identity gives $M_m(t) = [s^m]S_t(s)$ up to its finite inverse-row error.

Proof. Multiply numerator and denominator by $g(z^{-2}) + wz$ and retain the odd powers of z . The exact row-truncation identity in Lemma 2.5 supplies the error. The next lemma bounds this error on the whole closed domain, including $w = 0$. $\square$

Lemma 5.2 (The scattering approximation on the whole closed domain). *For each fixed polynomial background in the preceding small neighborhood, there are $C > 0$ and $0 < \kappa < 1$ such that*

$$\sup_{t \in \overline{D}} |M_m(t) - [s^m]S_t(s)| \leq C\kappa^m,$$

where S_t is the generating function in Proposition 5.1. The same bound holds for every Taylor coefficient after composition with f .

Proof. Use the scaled finite-section radius $\rho > 1$ from Proposition 2.7. Choose $1 < r_0 < R^2 < \rho$ inside its squared-variable collar. For t in the closed interior of $h(r_0\mathbb{T})$, the far-factor function F_w is holomorphic on $|z| < R^{-1}$ and uniformly bounded on that circle: any uncanceled physical zero has $|z| \leq \sqrt{r_0}$ in the original exterior variable. Thus $|f_j| \leq CR^j$. The row truncation argument now gives low-row errors bounded by $C(R/\rho)^N$. In the exact corner identity in Lemma 2.5, multiplication by the finitely many tail coefficients bounds the scattering error by $C(R^2/\rho)^N$.

This error is holomorphic in w on a fixed neighborhood of $\mathcal{W}$. For even N it is odd in w . Division by w therefore preserves an exponential bound on $\mathcal{W}$: near zero use Cauchy's estimate on a fixed small circle, and elsewhere divide by a fixed lower bound for $|w|$. With $N = 2m + 2$, the parity formula for S_t gives the assertion, including $t = -1$. The coefficient claim follows from the supremum bound on the unit disk after composition. $\square$

Lemma 5.3 (Coefficients of a Cauchy pole approaching the contact point). *Suppose $f : \mathbb{D} \rightarrow D$ extends to a $C^{1,\beta}$ diffeomorphism of the closed disk for some $\beta > 0$, is bi-Lipschitz, and satisfies $f(-1) = -1$, $f'(-1) > 0$. If $a \in D$ has $|a + 1| \leq C_0x$ and $\text{dist}(a, \partial D) \geq c_0x$, with $0 < x < x_0$, then*

$$\left| [u^k] \frac{1}{a - f(u)} \right| \leq Ce^{C_1 k x}.$$

The same estimate, multiplied by C/x , bounds the derivative of this coefficient with respect to a on a smaller such region. It also holds with a numerator $H(f(u))$, uniformly for a bounded family of functions H holomorphic on a fixed neighborhood of $\overline{D}$. The exponential constant C_1 depends only on the bi-Lipschitz bounds and C_0, c_0 , not on the Hölder seminorm.

Proof. Put $v = f^{-1}(a)$. The bi-Lipschitz bounds imply $|v + 1| \leq Cx$ and $1 - |v| \geq cx$. Remove the simple pole:

$$\frac{1}{a - f(u)} = \frac{1}{f'(v)(v - u)} + H_a(u).$$

The remainder is holomorphic in the disk and is bounded there by $C|u - v|^{\beta-1}$, by the Hölder Taylor remainder and the lower Lipschitz bound. Its boundary L^1 norm is uniformly bounded, so every Taylor coefficient is bounded by C . The pole term has coefficient $1/(f'(v)v^{k+1})$, bounded by $Ce^{C_1 k x}$. Cauchy's estimate in the parameter a , on a disk of radius cx contained in D , gives the derivative bound, with a possibly larger C_1 . To include a numerator, divide $H(t) = H(a) + (t - a)\tilde{H}_a(t)$; the divided differences are uniformly bounded and holomorphic near $\overline{D}$. $\square$

Proposition 5.4 (The growing conformal forcing bound). *For every fixed polynomial background in the preceding small neighborhood there are $\theta_0 > 0$ and C_P such that*

$$|[u^k]M_m(f(u))| \leq C_P m^{-3/2}, \quad 0 \leq k \leq \theta_0 m. \quad (11)$$

Moreover θ_0 can be chosen uniformly on a sufficiently small fixed $\mathcal{W}^{1+\alpha}$ neighborhood of the squared unit circle, for any fixed $\alpha > 0$. The constants C_P and the threshold for m may still depend arbitrarily on the fixed background. Consequently, for $-1/2 < s < 0$ and fixed $0 < \theta < \theta_0$,

$$\|P_{\lfloor \theta m \rfloor}(M_m \circ f)\|_{H^s} \leq C_{P,s} m^{s-1}.$$

Proof. Lemma 4.5 gives a $C^{1,\alpha}$ normalized map with $\|f' - 1\|_\infty < 1/4$ and uniform bi-Lipschitz bounds. Real symmetry gives $f(-1) = -1$ and $f'(-1) > 0$.

It is enough, by Lemma 5.2, to bound the double coefficient of $S_{f(u)}(s)$. Continue it from the s disk to a thin slit annulus $1 \leq |s| \leq 1 + \delta$, cut along $[-1 - \delta, -1]$. Its only nonanalyticity there

is the square root. The continued function $h(s^{-1})$ remains within $C\delta$ of Γ on this annulus: its derivative is continuous across the circle away from the cut and has a continuous limiting value at the contact point on both sides. By decreasing δ for the fixed background, the derivative bound here can be chosen absolute. Choose $R = 1 - L\delta$, with a fixed sufficiently large L depending only on the bi-Lipschitz constants of f . Then $f(\{|u| \leq R\})$ stays farther than $C\delta$ from Γ . No denominator zero is crossed in the slit annulus for these u .

Deforming the s coefficient contour gives an outer-circle term and an integral of the two boundary values on the cut. The outer term has u^k coefficient at most

$$C_P(1 + \delta)^{-m} R^{-k}.$$

This is exponentially small if $k \leq \theta_0 m$, with $\theta_0 < 1/(4L)$ and δ sufficiently small.

On the cut write $s = -1 - x$, $\zeta = -1/(1 + x)$, and

$$g_{\pm} = P(\zeta) \pm i\sqrt{x}, \quad t_{\pm} = \zeta g_{\pm}^2 - 1.$$

Since $P(\zeta) = O_P(x)$, these points satisfy

$$t_{\pm} = -1 + x + O_P(x^{3/2}), \quad |t_+ - t_-| \leq C_P x^{3/2}.$$

They and the segment between them lie in D at distance comparable to x from its boundary, after decreasing δ once more. The two numerators in S_t are $N_{\pm}(t) = \zeta A_t(\zeta) + \zeta^2 B_t(\zeta) g_{\pm}$. They are uniformly bounded holomorphic functions of t near $\overline{D}$, and $N_+ - N_- = O_P(\sqrt{x})$ in that norm. Split their jump as

$$\frac{N_+(t) - N_-(t)}{t_+ - t_-} + N_-(t) \left(\frac{1}{t_+ - t} - \frac{1}{t_- - t} \right).$$

Lemma 5.3, and its parameter derivative on the segment between t_- and t_+ , bound the u^k coefficient of this jump by $C_P \sqrt{x} e^{C_1 k x}$. This extraction is performed on a circle $|u| = R$ smaller than both pole radii, as ensured by the choice of L . The cut contribution is therefore bounded by

$$C_P \int_0^{\delta} x^{1/2} (1 + x)^{-m-1} e^{C_1 k x} dx \leq C_P m^{-3/2}$$

when $k \leq \theta_0 m$ and $\theta_0 < 1/(4C_1)$. The small circle about the branch point contributes zero in the contour deformation, since the boundary values are bounded there for $|u| \leq R$. This proves (11).

Only the uniform bi-Lipschitz constants and the fixed small C^1 neighborhood enter the restrictions on θ_0 . Lemma 4.5 supplies these uniformly on the stated $\mathcal{W}^{1+\alpha}$ neighborhood. The slit-annulus width may shrink with the particular polynomial; the restrictions on θ_0 above do not depend on that width. Finally, summing $(k + 1)^{2s} m^{-3}$ over $k < \theta m$ proves the Sobolev estimate. $\square$

Corollary 5.5 (Explicit parameters). *In Proposition 4.7 one may use $\gamma = 2^{-1000}$ and bound both operator norms and both inverse norms by 2^{100} . The forcing proportion may be taken as $\theta_0 = 2^{-20}$. Thus*

$$s = -\frac{1}{4}, \quad \tau = \frac{1}{8}, \quad \sigma = \frac{3}{8}, \quad \theta = 2^{-10000}$$

satisfy the strict oversampling inequality, with left side less than $1/2$. These choices are intentionally inefficient.

Proof. We record a loose constant ledger to make the parameter choices checkable. Use the coefficient norms already specified, and the periodic Hölder norm $\|v\|_{\infty} + \|v'\|_{\infty} + [v']_{\alpha}$ for $C^{1,\alpha}$. All the following estimates apply at $r = 3/4, 7/8$ and $\alpha = 1/8$.

In the endpoint lemma, $A_n \leq (n+1)^{-1/2}$ gives Schur row sums at most 24 and column sums at most 28, by splitting the displayed sums at $j = k+1$. Thus $\|E\| \leq 64$ is valid. The triangular inverse has squared Hilbert–Schmidt norm at most $\sum_{\ell \geq 1} \ell^{-3/2}(1 + \log \ell) < 8$, so its norm is at most 8. In the Cauchy-projection lemma the L^2 difference is at most 4δ ; differentiating gives at most 14δ in the equivalent derivative norm. Interpolation and the Fourier weight comparison give 64δ on H^r . The off-diagonal smoothing constant can be taken as 16 by its two Neumann resolvent factors.

The product estimates in the proof above give, successively,

$$\|(h-s)_+\|_{\mathcal{W}^{r+1}} \leq 20\gamma, \quad \|h' - 1\|_{\mathcal{W}_{\text{pos}}^r} \leq 21\gamma, \quad \|R - 1\|_{\mathcal{W}_{\text{pos}}^r} \leq 5\gamma, \quad \|m - 1\|_{\mathcal{W}_{\text{pos}}^r} \leq 52\gamma.$$

For example, the three positive terms in $h-s$ cost at most $16\gamma + 12\gamma^2$. The direct, positive-density, and negative-density errors in $B-I$ are bounded by 3328γ , 16384γ , and 40960γ respectively. The still looser bound $2^{25}\gamma$ therefore suffices.

The remaining norm comparisons can also be bounded generously. For the two values of r , the elementary sine inequalities give

$$n^{2r} \leq \int_{-\pi}^{\pi} \frac{|e^{int} - 1|^2}{|e^{it} - 1|^{1+2r}} dt \leq 64n^{2r} \quad (n \geq 1).$$

Splitting the upper integral at $1/n$ proves the upper bound; integrating only up to $1/n$ proves the lower bound. Thus fractional composition by a circle map with both metric and Jacobian bounds at most 2 costs less than 2^{20} . For $a = -2s \in \{1/2, 1/4\}$ the monomial beta weight lies between $\frac{1}{2}(k+1)^{-a}$ and $8(k+1)^{-a}$: for the lower bound integrate over $0 \leq 1-t \leq 1/(k+1)$, and for the upper use $(1-x)^k \leq e^{-kx}$. Analytic multiplier norms consequently cost at most 8 times the supremum norm.

For the conformal coordinate, radial conversion gives $\|\ell\|_{C^{1,\alpha}} \leq 2^{12}\gamma$. This follows by first taking the logarithm of $h(e^{it})e^{-it}$ and then inverting its angle, whose derivative is between $1/2$ and $3/2$. The cotangent-kernel proof bounds the Hilbert transform in the two required Hölder norms by 2^{20} . In the fixed-point proof one may take $M = 2^{24}\|\ell\|_{C^{1,\alpha}}$; composition costs at most $6\|\ell\|_{C^{1,\alpha}}$ on this set. It follows that $\|f' - 1\|_{\infty} \leq 2^{40}\gamma$ for $\gamma \leq 2^{-50}$. In particular all the metric and Jacobian bounds used above are at most 2 for the chosen γ .

The semigroup proof gives $\|b_x b_y / (b_x + b_y)\|_{\mathcal{W}} \leq 8$ and the associated base two-variable kernel norm at most 20. The triangular-indicator argument has bound $4n(1 + \log n) \leq 16n^{1+\alpha}$. Thus 2^{16} bounds both divided-difference constants needed here, including positive weights up to order 2. Neumann series, the logarithm and exponential, the derivative coefficient count, and the path of length at most $2|w|$ then bound the weighted sum for $A_t - 1, B_t$ by $2^{30}\gamma$.

Combining the preceding bounds in the explicit factorization, both bare norms are below 2^{40} , and the general differentiated density bound is below 2^{45} times its multiplier norm. The actual perturbation, including all coefficient multipliers, is below $2^{100}\gamma$. Its product with the bare inverse is therefore less than $1/2$ at $\gamma = 2^{-1000}$. The two actual norms and inverse norms are consequently below 2^{100} .

Finally the forcing proof can use $R = 1 - 32\delta$ for its coefficient-extraction circle. After shrinking the collar for the fixed background its continued derivatives are bounded by 2. The moving cut poles satisfy $|f^{-1}(a) + 1| \leq 4x$; also after the parameter Cauchy estimate their coefficient growth can be bounded by $C_P e^{64kx}$. These choices permit $\theta_0 = 2^{-20}$. The collar width and prefactor may depend on the background, as before. Now

$$2^{200}(4 \cdot 2^{-10000})^{1/8} < \frac{1}{2}$$

proves the stated oversampling assertion. $\square$

Lemma 5.6 (Oversampling from two Sobolev bounds). *Suppose A is an isomorphism on H^s and $H^{s+\tau}$, with $\tau > 0$ and compatible inverses. Write $M_s = \|A\|_{H^s \rightarrow H^s}$ and $K_r = \|A^{-1}\|_{H^r \rightarrow H^r}$. If $q \geq h$ and*

$$\varepsilon = M_s K_{s+\tau} \left(\frac{h}{q+1} \right)^\tau < 1,$$

then $P_h A P_q$ has a right inverse with H^s norm at most $K_s/(1-\varepsilon)$, whose output has degree less than q . No small bound for $A - I$ is assumed.

Proof. On $P_h H^s$, set $E = P_h A(I - P_q)A^{-1}P_h$. The tail estimate $\|(I - P_q)v\|_{H^s} \leq (q+1)^{-\tau} \|v\|_{H^{s+\tau}}$ and the finite-band estimate $\|P_h v\|_{H^{s+\tau}} \leq h^\tau \|P_h v\|_{H^s}$ give $\|E\| \leq \varepsilon$. Since $P_h A P_q A^{-1} P_h = I - E$, the operator $P_q A^{-1} P_h (I - E)^{-1}$ is the required right inverse. $\square$

Proposition 5.7 (Two explicit analytic estimates would suffice locally). *Fix one polynomial background P as above and numbers $0 < \theta < \sigma < 1/2$, $-1/2 < s < 0$, and $\tau > 0$. Let f be the exact conformal map onto D , with $f(0) = 0$ and $f'(0) > 0$, and define on polynomials*

$$\mathcal{A}_g x = (\mathcal{J}_g((J^0)^{-1}x)) \circ f.$$

Assume the following two estimates, established above on the specified small neighborhood and with the explicit parameters.

1. $\mathcal{A}_g$ extends to compatible isomorphisms on H^s and $H^{s+\tau}$, with

$$\|\mathcal{A}_g\|_{H^s} \|\mathcal{A}_g^{-1}\|_{H^{s+\tau}} (\theta/\sigma)^\tau < 1.$$

2. For $h = \lfloor \theta m \rfloor$, the exact forcing satisfies

$$\|P_h(M_m \circ f)\|_{H^s} \leq C_P m^{s-1}.$$

Then, for all sufficiently large m , real corrections supported at g_{-j} , $m - \lfloor \sigma m \rfloor + 1 \leq j \leq m$, make the first h jets vanish exactly. Their preconditioned vectors satisfy $\|x\|_{H^s} = O_P(m^{s-1})$, so all the coefficient and geometry conclusions of Corollary 3.3 hold.

Proof. Lemma 5.6 gives uniformly bounded right inverses for the first h rows and $q = \lfloor \sigma m \rfloor$ columns of the preconditioned limiting operator. By Corollary 2.9, the actual finite linear maps differ in H^s operator norm by an exponentially small quantity times a fixed power of m . Indeed (7) bounds the input coefficient norm by $C_s q^{-s} \|x\|_{H^s}$, while h uniformly bounded output coefficients have H^s norm at most $C_s h^{s+1/2}$ times their maximum. A Neumann correction therefore supplies uniformly bounded right inverses B_m for the finite maps.

Write the exact finite equations as $b_m + A_m x + N_m(x) = 0$. Corollary 2.8 and the same norm conversions bound the nonlinear remainder, and its difference on a ball, by an exponentially small quantity times fixed powers of m and the corresponding quadratic expression in x . The difference estimate also follows directly by subtracting the convergent separated-corner resolvent series. On a ball of radius Cm^{s-1} , the map

$$x \longmapsto -B_m b_m - B_m N_m(x)$$

therefore maps the ball into itself and has Lipschitz constant less than $1/2$ for large m , after increasing the fixed C . Its fixed point solves the exact equations. Since the background and the normalized conformal map have real coefficients, all these maps preserve real coefficient sequences, and the fixed point is real. The odd-pencil identity turns vanishing jets into the asserted characteristic-polynomial divisibility. Invisible restoration and the preceding corollary give the final bounds. $\square$

6 Exact selection and the infinite symbol

The following gives an exact selection rule. Its purpose is to remove any dependence on an unspecified choice of a nonlinear root. The algorithm is not intended for practical computation.

Theorem 6.1 (A continuous counterexample). *There is a computably specified continuous complex scalar symbol a with neither separate one-sided annular holomorphic extension and an increasing sequence n_j such that $T_{n_j}(a)$ has both 1 and -1 as eigenvalues of algebraic multiplicity at least $\lfloor \theta(n_j-1)/2 \rfloor$, where $\theta = 2^{-10000}$. Both points are outside $a(\mathbb{T})$. In particular its empirical eigenvalue measures do not converge to a_*m .*

Proof. Fix

$$\gamma = 2^{-1000}, \quad \theta = 2^{-10000}, \quad \sigma = \frac{3}{8}, \quad m_0 = 2^{10000}, \quad \kappa = 2^{-1010}, \quad \beta_j = \gamma 2^{-j-4}.$$

Start with $P_- = P_+ = 0$. At stage j , with the last chosen order m_{j-1} , first add the positive packet

$$Q_j(s) = \tau_j s^{m_{j-1}}(1+s), \quad \tau_j = \kappa 2^{-\lceil \sqrt{2m_{j-1}+1} \rceil}.$$

We will always choose $m_j \geq 8m_{j-1}$, divisible by eight. The new packet vanishes at -1 .

All positive costs fit their stage budgets. Its $\mathcal{W}^4$ norm is bounded by $C\tau_j(1+m_{j-1})^4$. The same bound, with a fixed larger constant, bounds its change to the squared curve in $\mathcal{W}^{1+\alpha}$, $\alpha = 1/8$. Indeed $g_0(1+s)$ belongs to that algebra, and the old finite polynomial has support of magnitude at most $2m_{j-1}+1$ and bounded $\mathcal{W}^0$ norm. Expand the square and use the weighted convolution inequality. The constant 2^{100} is more than sufficient in these estimates. For $x \geq m_0$ the function $(1+x)^{42-\sqrt{2x+1}}$ decreases and its value at $8x$ is less than a quarter of its value at x . Its initial value is smaller than the required bound for β_1 by the displayed choices. Thus each positive $\mathcal{W}^4$ cost and squared-curve cost is strictly less than β_j , uniformly in the subsequent choices of orders.

Hold the resulting background P fixed. For each prospective $m \geq 8m_{j-1}$ divisible by eight put $q = 3m/8$ and $h = \lfloor \theta m \rfloor$. Let $v = (v_0, \dots, v_{q-1})$ be real and set

$$L_v(s) = s^{-m} \sum_{d=0}^{q-1} v_d s^d, \quad D_v(s) = L_v(s) - L_v(-1)K_m(s).$$

Use the exact K_m in Lemma 3.1. Require the first h coefficients in $t = w^2 - 1$ of

$$\frac{\det(wI - T_{2m+1}(z(g + L_v)(z^2)))}{w}$$

to vanish. These are h real polynomial equations in the q variables v_d . There is no division ambiguity: the odd Toeplitz symbol makes the characteristic polynomial odd, and its quotient by w is a polynomial in w^2 , hence in t . The coefficients of these equations are computable from the current finite background and the rational binomial coefficients.

Put $x = J^0 v$ and impose also the strict inequalities

$$R_1 = \sum_{k < q} \frac{|x_k|}{k+1} < m^{-19/16}, \quad R_{1/2} = \sum_{k < q} \frac{|x_k|}{\sqrt{k+1}} < m^{-15/16}.$$

Require, separately, each of the following increments to be strictly below β_j :

$$\|D_v\|_{\mathcal{W}^0}, \quad \|D_v/g_0\|_{\mathcal{W}^0}, \quad \|g_0 D_v\|_{\mathcal{W}^{1+\alpha}}, \quad \|s(g + D_v)^2 - sg^2\|_{\mathcal{W}^{1+\alpha}}.$$

The last norm may be certified by the triangle bound obtained from $2sg_0D_v + 2sPD_v + sD_v^2$.

These conditions have regular solutions for all sufficiently large m . The two-level theorem, forcing bound, and oversampling lemma give the local completion with $\|x\|_{H^{-1/4}} = O_P(m^{-5/4})$. Thus $R_1 = O_P(m^{-5/4})$ and $R_{1/2} = O_P(m^{-1})$, which satisfy both strict bounds eventually. Conversely the stated strict bounds alone imply, by the restoration estimates,

$$\|D_v\|_{\mathcal{W}^0} = O(m^{-15/16}), \quad \|g_0D_v\|_{\mathcal{W}^{1+\alpha}} = O(m^{-1/16}), \quad \|D_v/g_0\|_{\mathcal{W}^0} = O(m^{-3/16}\sqrt{\log(m+1)}).$$

For the third bound use $H^{3/4}(\mathbb{T}) \subset \mathcal{W}^0$. The squared-curve increment is bounded by $C_P(m^{-1/16} + m^{-3/4})$. Hence the open norm budgets hold eventually as well.

The polynomial equations have full row rank at the supplied solutions. In the local completion proof their Jacobian differs by an exponentially small operator from a map with uniformly bounded right inverse. At a zero jet, multiplication by $\det(H - tU)$ and composition with f induce invertible triangular maps on the first h jets. The original polynomial equations therefore have the same full row rank.

Here is the deterministic selection rule. Enumerate all finite rational encodings consisting of an admissible m , a subset of h coordinates, rational values for the other coordinates, a rational cube $B(c, r)$ in the selected coordinates, and a rational square matrix K . If F is the resulting square polynomial system, seek strict interval certificates

$$\sup_{B(c, r)} \|I - KDF\|_\infty < \frac{1}{2}, \quad \|KF(c)\|_\infty < r/2.$$

Also certify both R bounds and all four norm budgets throughout the cube. To fix the order completely, write each rational in reduced form with positive denominator, list the coordinate subset in increasing order, and list matrix entries in row order. The height of an encoding is the maximum of m and the absolute values of all its integer numerators and denominators. In round N inspect all encodings of height at most N in lexicographic order, with absolute coefficient-enclosure error at most 2^{-N} , and select the first one whose strict certificates succeed. Each round is finite. Use tail-enclosure tolerance at most $\min(2^{-N}, \beta_j/8)$. The displayed estimates make $z \mapsto z - KF(z)$ a contraction of the cube into itself. They also imply that K is invertible. Its unique fixed point is the selected exact root, and the contraction iteration specifies that root as a computable real vector.

The coefficients remain computable by induction. The explicit stage budgets also give a computable uniform modulus of convergence for the final Fourier series. All norm certificates are computable. For a restored polynomial of maximum degree magnitude N_0 , the exact equality $D_v(-1) = 0$ and the binomial recurrences give, for $n > 2N_0$,

$$|[s^{-n}]g_0D_v| \leq 8N_0\|D_v\|_{\mathcal{W}^0}n^{-5/2}, \quad |[s^{-n}]D_v/g_0| \leq 8N_0\|D_v\|_{\mathcal{W}^0}n^{-3/2}.$$

Finite sums and integral tail bounds therefore give upper enclosures of the two infinite norms. More explicitly, past $J > 2N_0$ their respective omitted norm tails are at most $128N_0\|D_v\|_{\mathcal{W}^0}J^{-3/8}$ and $16N_0\|D_v\|_{\mathcal{W}^0}J^{-1/2}$. The constants follow from $A_n \leq (n+1)^{-1/2}$ and the first-difference recurrences for A_n and $\binom{1/2}{n}$. Use an enclosure tolerance less than a quarter of the stage budget. The remaining terms are finite polynomial coefficient norms. The cancellation in D_v is structural and holds throughout every cube, so this tail estimate does not use an approximately zero endpoint value.

The search terminates. At a regular root supplied above some h columns of the Jacobian form an invertible matrix. By the implicit-function theorem the other coordinates may be chosen rationally nearby while preserving a root and the strict budgets. A sufficiently small rational cube and a sufficiently accurate rational inverse preconditioner satisfy both contraction inequalities. Refining

the coefficient and tail enclosures eventually certifies all the strict conditions. The dovetailed search reaches such a certificate in a finite round.

Choose this m as m_j , add its exact D_v to P_- , and continue. The sums of the positive and negative stage costs are smaller than $\gamma/8$. Thus all finite backgrounds remain strictly within the neighborhood of the operator theorem: the three negative norms and the positive $\mathcal{W}^4$ norm are controlled separately, and both types of squared-curve increments have summable bounds. This proves the induction.

The resulting series converges in $\mathcal{W}^0$ to a continuous g , hence to a continuous $a(z) = zg(z^2)$. Every later change is invisible to a previously selected $T_{2m_j+1}(a)$. The next positive packet starts at Fourier frequency $2m_j+1$ in a ; the next negative correction starts beyond $4m_j$ in the g variable; and the current restoring packet starts at g_{-m_j-1} . All lie outside the diagonals used by the previously fixed section. Its exact characteristic factor $(w^2 - 1)^{\lfloor \theta m_j \rfloor}$ therefore persists in the final symbol.

The positive coefficient at g_{m_j-1} is exactly τ_j , and $\log \tau_j / (2m_j+1) \rightarrow 0$. At g_{-3m_j} no correction ever acts: the current restoration ends at $-2m_j - 1$ and the next correction starts beyond $-4m_j$. These coefficients are the original nonzero binomial coefficients, of order $m_j^{-3/2}$. Consequently both positive and negative Fourier root limsups of a equal one. An outer annular extension would give $|a_k| \leq C_\rho \rho^{-k}$ for some $\rho > 1$ by Cauchy's Laurent coefficient formula; an inner extension would give the analogous geometric decay of a_{-k} . Both are excluded.

Finally $a_0(\mathbb{T})$ satisfies $|w^2 - 1| = 1$ and its distance from either 1 or -1 is $\sqrt{2} - 1$. For example, $1 = |w - 1||w + 1| \leq d(2 + d)$, where $d = |w - 1|$, and equality is attained at $w = \sqrt{2}$. The total uniform perturbation is smaller than 2γ , so both points remain outside $a(\mathbb{T})$. The explicit test function $\Phi(w) = \max\{0, 1 - 8|w - 1|\}$ is continuous, nonnegative, compactly supported, and zero on $a(\mathbb{T})$. Its integral against a_*m is zero. At $n_j = 2m_j + 1$ its empirical eigenvalue integral is at least $\lfloor \theta m_j \rfloor / (2m_j + 1) \rightarrow \theta/2 > 0$. This proves the asserted failure of canonical convergence. $\square$

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